

Under Pressure: Securing Water in Global Security (SWIGS)

NATO Science for Peace and Security Call for Proposals 2026-1

Multi-year Project – Phase 1 Concept Note

1. Security Challenges Addressed by the Project

Water underpins essential societal functioning and logistical support for military operations. When its supply, availability, or quality are contested or compromised, civilians and military personnel are gravely affected. The “Securing Water in Global Security” (SWIGS) project addresses the risks that water-related violence and conflict pose to NATO’s strategic interests and operations to secure lasting peace and stability. As pressures on water resources and infrastructure mount, intersecting with complex environmental, economic, and social risk factors, this project aims to better assess and prepare for water-related security risks within and beyond Allied borders. Violence and conflict involving water resources are critical, under-assessed, and under-addressed pieces of the cascading crises of conflict and climate change across regions of strategic importance to NATO and its partners.

Competition over rivers and aquifers, damage to water supply, treatment, and storage infrastructure, inequitable access to water, sanitation and hygiene, and climate-driven hydrological extremes, including droughts and floods, have long contributed to local, national, and transboundary tensions. Civilizations have collapsed due to prolonged drought. Contested access to and control of water and food supplies, and conflict over access to and control of decision-making for managing these vital resources has triggered violence. Water and agricultural systems have been targeted during wars and used as weapons of warfare.

Today, these risks are at their highest level as climate change accelerates hydrologic variability, populations grow, urbanization expands, and critical civilian infrastructure becomes more interconnected and vulnerable. Violent conflicts rage around the world, devastating civilian infrastructure and the environment. On NATO’s direct borders, extensive damage to water supplies in Ukraine and Syria has displaced millions of civilians and eroded social cohesion. Recent hybrid threats such as Russian, Iranian, and Chinese-linked cyber-attacks of municipal water in Canada, Denmark, Norway, the United Kingdom, and United States are testing the capabilities of NATO members and partners to safeguard water. The international team for the project holds leading expertise in water-conflict data compilation and assessment, theory and practice around water weaponization and climate insecurity, policy-relevant analysis, Euro-Atlantic and Indo-Pacific regional expertise, lived experience of conflict, and direct engagement with defense and security communities.

Causes of water-related violence must be understood, critical civilian water infrastructure made more resilient, and solutions found and implemented. The proposed project directly addresses NATO Science for Peace and Security (SPS) Programme priorities on **Environment, Climate Change and Security** by strengthening the capability of NATO member and partner countries to anticipate, understand, and respond to water-related security risks before they escalate into violence or humanitarian crises. The project also advances NATO priorities on **Defense against Hybrid Threats** and **Resilience** by evaluating recent experience related to the vulnerability of critical civilian water infrastructure in peacetime and conflict, and development of strategies and institutional systems to improve the resilience of that infrastructure. By better understanding how hybrid tactics and hydrologic extremes contribute to violence today, military and civilian populations can more effectively protect water resources and infrastructure in resilient, holistic ways.

In alignment with **Human and Social Aspects of Security**, the project explicitly incorporates social, political, and economic drivers of conflict—recognizing that water-related violence rarely stems from physical scarcity alone. Governance failures, inequitable access and allocation, and marginalization of vulnerable populations all shape conflict dynamics. The proposed project integrates these dimensions to provide a more nuanced and realistic understanding of security risks. The work will also include an assessment of climate drivers on the vulnerability of water systems and water users.

The project aligns with **Operational Support** by delivering actionable, decision-relevant intelligence products for policymakers, defense planners, and civilian stakeholders. These products will enhance strategic foresight, regional risk assessment, and scenario planning relevant to NATO missions, partner-country stability efforts, and civil-military coordination in contested or climate-stressed environments.

2. Technical Challenges Addressed by the Project

The project will address three technical challenges aligned with NATO's security priorities:

- **Challenge 1: Transitioning to AI-Assisted Data Retrieval and Categorization.** We will transform the Pacific Institute's Water Conflict Chronology from a manual, retrospective dataset into a continuously updated, AI-assisted, and semi-autonomous data retrieval and categorization system. This shift enables near real-time monitoring of water-related violence, targeting of infrastructure, and resource weaponization, allowing for more immediate identification of water-related conflicts, hotspots, and trends.
- **Challenge 2: Determining Drivers and Delineating Resilience.** We will investigate the drivers of water-related violence and strategies for reducing vulnerability and increasing the resilience of water infrastructure. This includes an assessment of the role of climate change and extreme events as contributing triggers for water-related violence, and the vulnerability of military operations and facilities to climate extremes. Identifying the most pressing hybrid threats to water will also be a focus. Altogether, this analysis will help to better understand the underlying causes of water-related violence and will help parse out the nuances between different types of water violence and conflict pathways.
- **Challenge 3: Operationalizing Data for Early Warning.** Through an Advanced Research Workshop, we will explore the value and potential to use historical and near real-time water data in conflict forecasting tools and approaches. For groups attempting to develop early warning, we will contribute expertise around data availability and policies of resilience and risk to help ensure that technical insights from security experts and analysts can be translated into actionable early-warning systems for the NATO security community.

3. State-of-the-Art and Existing Research and Data

The U.S.-based Pacific Institute's Water Conflict Chronology is the world's most comprehensive open database on water-related violence, with more than 2,750 documented events. The database has been widely used in defense and intelligence assessments, scholarly analyses, and public outreach and information sharing, and these data will be central to the analysis in the project. The Center for Strategic and International Studies in Washington, DC complements this empirical foundation with policy-relevant analysis to define the future of national security, regional expertise, and direct engagement with defense and security communities. Partnership with the Australian National University's Department of International Relations in the Coral Bell School of Asia Pacific Affairs brings expertise in conceptualizing and restraining water weaponization, and the use of data for early warning and strategic foresight priorities of NATO. Ukrainian partners in Kyiv and Kherson bring experience in real-time vulnerabilities of urban water infrastructure and building resilience of water

systems in the face of conflict. Together, this cross-regional partnership will strengthen coordination on local and non-geographically bound challenges for water in today's global security landscape and deepen qualitative and quantitative assessment of water-security linkages.

4. Project Deliverables & Advancement of State-of-the-Art

The project advances water security analysis by linking AI-enabled monitoring, empirical case studies, and practitioner engagement. Near real-time identification of water-related conflict is combined with in-depth data analysis to understand conflict drivers and case studies—particularly from Ukraine—to assess infrastructure vulnerability and resilience in challenging environments. An Advanced Research Workshop translates these findings into decision-relevant insights for NATO's strategic foresight and resilience planning.

1) Research

Task 1 – Improve Tools for Collecting and Analyzing Data on Water-Related Conflicts: Water Conflict Chronology Database Update and Expansion: Deploy and validate AI-assisted classification workflows that enable scalable, near real-time identification of water conflict events from news sources, government reports, and armed conflict databases. This represents a methodological advance over manual and fully automated approaches by combining speed, transparency, and expert validation.

Deliverables

- Update the Water Conflict Chronology comprehensive database
- Develop an AI classification model for ongoing, semi-autonomous updates to the Water Conflict Chronology, informed by a clear delineation of water conflict categories and coding criteria
- Explore methodologies for reviewing AI-classified water conflict data, drawing on specialist coding expertise
- Report on data and AI model validation

Task 2 – Data Analysis – Trends and Drivers: Conduct analysis to understand contemporary trends and principal drivers underlying water-related violence and conflicts. Focus will be placed on hybrid threats, as well as environmental drivers, including climate change impacts and extreme events, as a force multiplier. The analysis will distinguish between proximate triggers and underlying structural drivers, and mediating socio-economic and governance factors, improving the ability to assess which water-related risks are most likely to escalate into violence.

Deliverables

- Prepare a report on trends in water-conflict drivers, including climate change drivers
- Prepare a brief on insights derived from data analysis

Task 3 – Ukraine and Other Case Studies will be prepared that inform key security risks relevant to NATO countries in military, economic, and political terms, with a focus on regions where water-related violence is particularly endemic. Of particular interest is Ukraine where urban water systems and related energy systems have been extensive casualties of the Russia-Ukraine War. As the availability of non-classified information permits, the Ukrainian co-investigators will prepare a detailed assessment of their experience with attacks on urban water and related energy systems, and the tools and strategies that they have developed to build more resilience systems while under fire. These case studies will generate transferable lessons for protecting civilian

water infrastructure in acute crises, with direct relevance to military planning, civil defense, and post-conflict recovery.

Deliverables

- Prepare case study reports and briefings.

2) Generate Actionable Decision Support

Task 4 – Assess the Role that Water-Conflict Data can play in the potential development of strategic foresight tools. We will hold an Advanced Research Workshop (ARW) that brings together practitioners who are developing or using forecasting or early-warning tools to share information on the strengths, weaknesses, and development needs for more successful tools. The workshop serves as the primary mechanism for translating technical findings into formats and concepts usable by NATO analysts, planners, and partner-country institutions. We will summarize the lessons from the workshop in a short white paper, and socialize these with policymakers to integrate them into Euro-Atlantic peacebuilding efforts.

Deliverables

- Develop and hold an Advanced Research Workshop
- Prepare an ARW summary white paper

Task 5 – Outreach and Communication Efforts and Tools

Develop reports and outreach that translates research findings into actionable insights for policymakers.

Deliverables

- Prepare a policy brief with actionable recommendations
- Present briefings to relevant policy makers
- Prepare a peer-reviewed journal article

5. Methodology

1. AI-enabled identification and classification of water-related conflict

The project will apply a lightweight artificial intelligence text-classification model trained on a curated subset of historical water conflict events. The model will scan open-source reporting (e.g., news, public reports) to identify potential water-related conflict events and assign them to conflict categories (e.g., trigger, casualty, weapon). All AI-identified events will be subject to specialist human review, ensuring data quality, transparency, and analytical reliability while significantly improving timeliness and scale compared to fully manual approaches.

2. Analysis of conflict drivers and risk multipliers

Quantitative and qualitative analyses will be conducted to examine the environmental, political, economic, and governance drivers associated with water-related conflict. Special attention will be paid to hybrid threats, climate-driven hydrologic extremes (droughts, floods), infrastructure vulnerability, and governance stressors that act as force multipliers for instability. This analysis will support identification of patterns relevant to NATO's strategic foresight and resilience planning.

3. Resilience-focused case studies

Targeted case studies, including Ukraine and other strategically relevant regions, will assess how attacks on or failures of civilian water infrastructure are affecting civilian lives and military operations, including economic and governance impacts. These case studies will emphasize practical lessons learned and resilience strategies applicable to NATO member and partner countries.

4. Integration with strategic foresight and early-warning practice

An Advanced Research Workshop will convene experts from defense, policy, and research communities to evaluate how historical and near-real-time water-conflict data can be incorporated into existing risk assessment, forecasting, and early-warning frameworks. The workshop will focus on operational relevance rather than tool development, ensuring outputs are decision-support oriented.

6. Project KPI's

KPI's measurable within the project timeline

- Increase in data quality, completeness and timeliness of documented water-related conflict events in the Water Conflict Chronology
- Demonstrated accuracy and reliability of AI-assisted classification relative to expert human review
- Reduction in staff time required to identify and code conflict events relative to historical baselines
- An assessment from Ukrainian experience of the vulnerability of civilian water systems under fire in conflict zones, and strategies for improving resilience of those systems.
- Successful design and delivery of the Advanced Research Workshop, including participation from NATO member and partner country experts
- Completion and dissemination of a final policy-focused briefing with actionable recommendations relevant to NATO security priorities

Measurable beyond the project timeline

- Evidence of uptake or citation of project findings in NATO-related policy, planning, or analytical contexts
- Use of water conflict insights to inform resilience planning, infrastructure protection, and/or strategic foresight efforts of partner countries

7. Project Impact

The project will strengthen NATO and partner countries' capacity to anticipate, understand, and respond to water-related security risks by improving access to timely, credible, and policy-relevant data and case insights on water-related conflict. By linking environmental stressors, infrastructure vulnerability, and conflict dynamics, the project supports more integrated risk assessment, resilience planning, and civil-military coordination. Over time, the project's outputs will contribute to earlier identification of emerging instability, better protection of critical civilian water infrastructure, and more informed decision-making in climate-stressed, conflict-affected and hybrid security environments of strategic importance to NATO.